Ricardo Jesus Huaman Kemper

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Research Interests

Robot Learning, Computer Vision, Natural Language Processing, Deep Reinforcement Learning

Education

Antenor Orrego Private University (UPAO), Trujillo, Peru

B.E. Electronic Engineering **July 2017 – July 2022**

- GPA: 15.86/20, Class Rank: 1st

Research Experience

Cardiff University, Cardiff, UK

Research Assistant, Human-centred Computing Laboratory **January 2024 – April 2024**

- Advisors: Dr. Carolina Fuentes & Dr. Fernando Alva
- Through the REPUcs 2024 Program, led the integration of foundational AI models with a humanoid robot, enhancing human-robot interaction in educational environments
- Implemented large language models (LLMs) into a quadruped robot simulation for autonomous indoor navigation, enabling more adaptive and intelligent behaviors in complex environments

Antenor Orrego Private University (UPAO), Trujillo, Peru

Research Assistant, Multidisciplinary Research Laboratory **May 2021 – December 2023**

- Advisors: Dr. Sixto Prado & Clayder Gonzalez, MSc
- I worked on the development of an agricultural mobile robot's autonomous navigation system, incorporating SLAM, path planning, and motion control algorithms. The components were integrated using the ROS middleware. This project aimed to enhance automation and productivity in the agricultural sector. Supported by government grant 171-2020-FONDECYT
- I worked on creating computer vision pipelines using deep learning to address specific agricultural challenges. Some examples include artichoke seedling classification, pre-harvest blueberry ripeness categorization, and the detection of missed seeding in plug trays. Supported by government grant 79170-2022-PROCIENCIA

Selected Publications

[1] A. Alamo, **R. J. Huaman Kemper** and S. R. Prado Gardini, "Integrated Opposite-Phase Kinematics, Hybrid-A* Planning, and MPPI Control for Autonomous Navigation in Agricultural Environment," 2026 UR, Ibaraki, Osaka, Japan, 2026, pp. 468-475, doi: 10.1109/UR69537.2026.11626756

[2] **R. J. Huaman Kemper**, P. A. Zamora Manrique, D. D. Herrera Rosas and S. R. Prado Gardini, "Performance Evaluation of Detection and Tracking Algorithms for Automated Grape Cluster Counting," 2025 IEEE C3, 2025, pp. 1-6, doi: 10.1109/C366505.2025.11340275

[3] **R. Huaman**, C. Gonzalez and S. Prado, "Comparative Analysis of LiDAR Inertial Odometry Algorithms in Blueberry Crops," Engineering Proceedings, vol. 83, no. 1, 2025, doi: 10.3390/engproc2025083009

[4] **R. Huaman**, C. Gonzalez and S. Prado, "Performance Evaluation of the ROS Navigation Stack using LeGO-LOAM," BTSym'23, Campinas, Brazil, 2024, pp. 173-183, doi: 10.1007/978-3-031-66961-3_16

[5] P. B. Cubas Muñoz, **R. J. Huaman Kemper** and S. R. Prado Gardini, "Artificial vision strategy for Ripeness assessment of Blueberries on Images taken during Pre-harvest stage in Agroindustrial Environments using Deep Learning Techniques," 2023 IEEE INTERCON, Lima, 2023, pp. 1-6, doi: 10.1109/INTERCON59652.2023.10326058

[6] P. B. Cubas Muñoz, **R. J. Huaman Kemper**, E. M. Fiestas Sorogastua and S. R. Prado Gardini, "Deep Learning-based Segmentation and Classification System for Artichoke Seedling Grading," 2023 IEEE INTERCON, Lima, 2023, pp. 1-6, doi: 10.1109/INTERCON59652.2023.10326060

[7] J. A. Bazan Quispe, **R. J. Huaman Kemper** and S. R. Prado Gardini, "Linear Quadratic Regulator (LQR) Control for the Active Suspension System of a Four-Wheeled Agricultural Robot," 2023 IEEE INTERCON, Lima, 2023, pp. 1-6, doi: 10.1109/INTERCON59652.2023.10326049

[8] **R. J. Huaman Kemper**, C. Gonzalez and S. R. Prado Gardini, "Autonomous Navigation of a Four-Wheeled Robot in a Simulated Blueberry Farm Environment," 2022 IEEE ANDESCON, Barranquilla, Colombia, 2022, pp. 1-6, doi: 10.1109/ANDESCON56260.2022.9989865

Industry Experience

Bank of Credit of Peru (BCP), Lima, Peru

Machine Learning Engineer — **October 2025 – Present**

- Co-built and optimized an internal framework for the continuous monitoring of 40+ production models spanning consumer, SME, and wholesale banking segments, safeguarding the operational reliability of business-critical models across the bank
- Migrated and managed 15+ analytical model pipelines into production for consumer banking under tight deadlines, powering bank-wide marketing campaigns that reach millions of clients
- Co-lead an internal Advanced Analytics COE knowledge-sharing program, coordinating monthly technical sessions that reach 100+ Machine Learning Engineers, Data Scientists, and Analytics Translators

Intelligent Platforms, Lima, Peru

Artificial Intelligence Engineer — **May 2024 – September 2025**

- Built object detection and real-time facial recognition systems for streaming media using gRPC for high-efficiency inter-service communication, cutting manual processing time by 60% and identifying over 1,000 individuals across multimedia content
- Designed a scalable, microservices-based infrastructure to deploy text classification and named entity recognition (NER) models, optimizing inference speed across 50+ models by applying MLOps best practices
- Developed LLM-powered RAG agents (LangChain, Agno, vector/relational databases) and an embedding-based thematic discovery pipeline, delivering context-aware responses and automated topic identification across large volumes of text

Academic Talks and Poster Sessions

R. Huaman, "Computer Vision: Applications in Academia and Industry". Talk, *IEEE Signal Processing Society Student Branch at UNMSM*. September, 2024

P. Cubas*, **R. Huaman*** and S. Prado, "Deep Learning Approach for Accurate Pre-Harvest Blueberry Ripeness Classification". Poster, *Workshop on Robotics in Agriculture at **IROS***. October, 2023

C. Gonzalez*, **R. Huaman*** and S. Prado, "Comparative Analysis of Lidar Odometry Algorithms in Blueberry Crops". Poster, *Workshop on Robotics in Agriculture at **IROS***. October, 2023

R. Huaman, C. Gonzalez, and S. Prado, "Integration of an Autonomous Navigation System for an Agricultural Mobile Robot in a Simulated Blueberry Cultivation Environment". Talk, *XIII International Scientific Meeting of the North at UPAO*. December, 2022

R. Huaman, C. Gonzalez, and S. Prado, "Implementation of an Autonomous Navigation System for a Mobile Robot in a Simulated Environment of a Blueberry Agro-Industrial Farm". Talk, *I International Congress of Research and Innovation Incubators at UPAO*. November, 2022

R. Huaman, "Path Planning for Autonomous Robots". Talk, *Electronics Engineering Dept. at UPAO*. May, 2022

Honors and Awards

First Place at the Scientific Article Competition — September 2025
CONEIMERA, UNT

Selected for the REPUcs 2024 Program — September 2023
Research Experience for Peruvian Undergraduates (REPU)

2nd Place at the Research Projects Competition — August 2023
CONEIMERA, UNSA

Service and Competitions

Team Member, European Rover Challenge 2025 (Remote Edition) — April – August 2025
Team Terrabots, autonomous navigation

Volunteer Staff, reviewed applications for the program — July 2025
Research Experience for Peruvian Undergraduates (REPU)

Academic Reviewing

IEEE International Conference on Electronics, Electrical Engineering and Computing (INTERCON): 2023

Programming & Software

Python, C++, PyTorch, TensorFlow, ROS 1 & 2, Gazebo, Git, Linux, MATLAB, AWS, Azure, Docker, Kubernetes, LangChain

Languages

Spanish (*native*), English (*TOEFL iBT 104/120*), Japanese (*JLPT N2*)